Biomedical Signal Processing and Control 100 (2025) 107167

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

Automated pericardium segmentation and epicardial adipose tissue quantification from computed tomography images

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ARTICLE INFO

Keywords:
Computed tomography
Deep neural network
Pericardium segmentation
Epicardial adipose tissue

ABSTRACT

Background and Objective: Epicardial Adipose Tissue (EAT) is regarded as an independent risk factor for cardiovascular disease, and an increase in its volume is closely associated with disorders such as coronary artery atherosclerosis. Traditional manual and semi-automatic methods for EAT segmentation rely on subjective judgment, resulting in uncertainty and unreliability, which limits their application in clinical practice. Therefore, this study aims to develop a fully automatic segmentation and quantification method to improve the accuracy of EAT assessment.

Methods: A Boundary-Enhanced Multi-scale U-Net network with a Convolutional Transformer (BMT-UNet) is developed to segment the pericardium. The BMT-UNet comprises Boundary-Enhanced (BE) modules, Multi-Scale (MS) modules, and a Convolutional Transformer (ConvT) module. The MS and BE modules in the encoding part are designed to capture detailed boundary features and accurately delineate the pericardium boundary by combining multi-scale features with morphological operations, leveraging their complementarity. The ConvT module integrates global contextual information, thereby enhancing overall segmentation accuracy and addressing the issue of internal holes in the segmented pericardial images. The volume of EAT is automatically quantified using standard fat thresholds with a range of −190 to −30 HU.

Results: For a Coronary Computed Tomography Angiography (CCTA) dataset which contained 50 patients, the Dice coefficient and Hausdorff distance for the proposed method of pericardial and EAT segmentation are 98.3% $\pm$ 0.2%, 5.7$\pm$0.8 mm, and 93.9% $\pm$ 1.7%, 2.1 $\pm$ 0.3 mm, respectively. The linear regression coefficient between the EAT volume segmented and the actual volume is 0.982, and the Pearson correlation coefficient is 0.99. Bland-Altman analysis further confirmed the high consistency between the automated and manual methods. These results demonstrate a significant improvement over existing methods, particularly in terms of segmentation precision and reliability, which are critical for clinical application.

Conclusions: This work develops an automated method for quantifying EAT in Computed Tomography (CT) images, and the results agreed closely with expert evaluations. Code is available at: https://github.com/wy-9903/BMT-UNet.

1. Introduction

Epicardial adipose tissue (EAT) is a type of fatty tissue located within the pericardium, forming a protective layer around the heart, mostly in the atrioventricular and interventricular grooves, and partially extending into the myocardial tissue [1]. It is usually distributed along the

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https://doi.org/10.1016/j.bspc.2024.107167
Received 28 December 2023; Received in revised form 29 October 2024; Accepted 2 November 2024
Available online 5 November 2024

outer membrane of the coronary artery branches and is anatomically closely connected to them [2]. EAT is vascularly connected to the myocardium and coronary arteries, enabling inflammation within the epicardial fat to directly impact the health of the heart and coronary arteries, thereby promoting the development of atherosclerosis [3]. EAT can promote the occurrence and development of cardiovascular disease by releasing pro-inflammatory factors, oxidative stress and endothelial damage [4]. Additionally, it can lead to local inflammation, fibrosis in the left atrium, and infiltration of adipose tissue, which can ultimately result in the onset of atrial fibrillation [5]. Several studies have demonstrated a significant independent correlation between epicardial adipose tissue volume (EATV), myocardial ischemia and coronary artery stenosis, as well as major adverse cardiovascular events [6,7]. EATV has emerged as a critical indicator in the assessment of cardiovascular disease [8]. Therefore, it would be desirable to be able to achieve accurate and non-invasive quantification of patients in clinical practice.

In measuring EATV, typical imaging methods include Ultrasound, Magnetic Resonance imaging (MRI) and Computed Tomography (CT) [9]. Among them, CT, with its fast-imaging speed, high resolution, high contrast and low noise, is currently the most commonly used non-invasive screening and diagnostic method for people at high risk of coronary artery disease, and quantitative studies of EAT are currently based on CT imaging [10]. However, due to the complexity of EAT distribution around the heart [11], manual segmentation and quantification of EAT is time consuming and imposes a high demand on doctors or technicians. In addition, manual segmentation methods rely on the subjective judgment and experience of the operator [12], which may lead to inconsistency and inaccuracy in segmentation results. Meanwhile, the development of automated analysis of medical image can alleviate the burden on clinicians and provide additional information for disease prevention and treatment [13,14].

With the advancement of image processing techniques, some researchers have proposed semi-automatic methods for the segmentation and quantification of EAT. Dey et al. [15] achieved quantification of EAT by identifying the top and bottom slices of the heart and manually selecting 5–7 control points to determine the position of the pericardium. Militello et al. [16] proposed a GUI tool that enables experts to quickly segment EAT. Their methods require an expert to define the Volume of Interest (VoI) on a few slices of the scan, and the rest of the VoI is then interpolated. In the quantification of EAT in 3D CT, Vladimir et al. [17] proposed a semi-automatic method that uses local adaptive morphology and fuzzy c-means clustering for slice-by-slice segmentation to segment and quantify EAT from 3D CT images. Although these semi-automatic methods can accelerate the segmentation and quantification process, they still require the participation of experts, leading to inter-observer variability. Hence, this is impractical for large-scale studies or clinical practices.

In recent years, deep learning-based methods have achieved success in various medical image segmentation tasks. U-Net, as a convolutional neural network architecture, is widely used in medical image segmentation tasks [18]. It consists of an encoder and a decoder, where the encoder extracts multi-level features of the image through successive convolution and pooling operations, while the decoder performs upsampling to restore the image resolution. Additionally, the skip connections between the encoder and decoder allow low-level features to be transferred directly to the corresponding layers in the decoder, thereby enhancing the model's feature extraction capability. Turk [19] proposed the RNGU-NET, which enhances the encoder by incorporating a ResNet structure into the U-Net architecture and introduces a Gate Attention Block (GAB) in the decoder to better extract important features. Additionally, to address the bottleneck issue in feature transmission within U-Net, an improved Non-Local Block (NLB) structure was designed to reduce feature loss. Milletari et al. [20] proposed V-Net, which employs a fully convolutional 3D architecture and introduces residual connections in both the encoding and decoding phases to better capture details and reduce information loss. Building upon V-Net, Turk et al. [21] introduced a two-stage bottleneck block structure to help minimize the loss of input information. However, while V-Net incorporates 3D spatial information, it results in higher computational costs. For small targets such as the pericardium, 3D convolutions may capture excessive background information, reducing segmentation accuracy. Furthermore, 3D models in medical imaging require large amounts of annotated data, which are often difficult to obtain and expensive to label. Based on these considerations, we chose to modify the U-Net architecture. Wherein, by implementing a weighted multi-slice strategy, we effectively approximate 3D features using 2D information, striking a balance between performance and computational efficiency.

Researchers have also started exploring deep learning techniques for achieving fully automated segmentation and quantification of EAT [22]. Li et al. [23] employed an enhanced U-Net model for the segmentation of EAT and pericardial fat. They incorporated pyramid pooling layers into the U-Net architecture to better understand and segment EAT regions of various sizes. Additionally, they utilized focal loss to address the issue of class imbalance between pericardial fat and background. A similar approach was proposed by Benčević et al. [24]. The difference is that they converted the traditional rectangular coordinate image into a polar coordinate representation and used the features in polar coordinates for training. To reduce training complexity and improve segmentation speed, Li et al. [25] first excluded non-adipose pixels by threshold-based segmentation, after which the resulting image was fed into a modified U-Net network. This approach incorporates a residual dilated convolution module at the bottom of the U-Net to enlarge the receptive field, effectively reducing the number of false-positive pixels. However, the introduced residual dilated convolution module may exhibit high sensitivity to large-scale features but low sensitivity to small-scale features (e.g., small adipose tissue regions). Consequently, the aforementioned studies have not fully leveraged multi-scale features, and the extracted single-scale features may fail to capture all useful information in the images. This limitation could result in inadequate segmentation of certain details. While most of the segmentation methods for EAT are based on 2D images, He et al. [26] proposed a 3D deep attention U-Net method on 3D patches extracted from CT volumes to automatically segment EAT. This method utilizes attention gates to focus on the boundary structure of the EAT and exclude noisy data from the training process. Recently, Zhao et al. [27] utilized SwinUNETR as the backbone network to extract features from the input three-dimensional images. By integrating the random feature vectors generated by the designed Probabilistic Net, they produced more precise segmentation results. Additionally, they designed a Discriminator Net to merge the coarse segmentation results with the ground truth, thereby enhancing the segmentation accuracy, particularly in capturing small lesion features. However, processing 3D images requires more computational resources than 2D images, which can result in longer training times and higher memory demands, thereby limiting its application in resource-constrained scenarios.

The boundary between EAT and the surrounding tissues is often blurred, and there are significant differences in the morphology, location, volume and density of the fatty areas among various populations and individuals, which makes it challenging to quantify the EATV. To solve this problem, some researchers have adopted an indirect approach by initially segmenting the pericardium and then quantifying the EATV within the pericardial contour. For example, the first application of convolutional neural networks (CNNs) in EAT segmentation was proposed by Commandeur et al. [28], where a two-stage convolutional network was used to segment the thoracic adipose tissue, and the pericardial contour was extracted by a statistical shape model, after which the total EATV was obtained by thresholding. However, their method lacks accuracy in the initial pericardial segmentation, particularly in cases where the pericardial boundary is unclear. Errors in the initial segmentation can propagate into the quantification of EATV, leading to inaccurate final results. To address this issue, morphological operations may be needed to further optimize the pericardial segmentation results, reducing noise and errors. Yang et al. [29] proposed a morphological vision transformer network, which integrates the dominant features extracted from the images and further refines these features through learning-based morphological operators, such as dilation and erosion. Zhang et al. [30] used a dual U-Net structure with a morphology processing layer to improve EAT segmentation. Their methods include two U-Net networks, the first U-Net aimed to detect the pericardium and segment its internal regions, following which the internal regions are refined by connecting a morphological processing layer. The second U-Net was used to search and segment EAT from the processed inner region of the pericardium obtained from the first network, obtaining more accurate EAT segmentation results with a mean Dice score of 0.9119 on a cardiac CT dataset of 20 patients [31]. Although their study achieved an improvement in Dice score by adding a morphological processing layer, it did not utilize multi-scale feature extraction. Relying solely on morphological operations may be limited by fixed shapes and scales, which can affect performance when handling morphologically diverse medical images. In order to better utilize the contextual relationships between slices, Li et al. [22] used a multi-slice deep neural network to simultaneously segment the pericardium in multiple adjacent slices, and then used a deformable model to reduce false positive and negative regions in the segmented binary pericardial images [32–35]. Finally, a ROI was defined through the pericardial mask, and the thresholding method was used to segment and quantify the EAT. This method takes advantage of multi-scale feature extraction, but compared to morphological operations, it may overlook some local details. Issues such as small holes in pericardial segmentation may not be adequately addressed.

Although these methods have improved the speed and accuracy of EAT quantification, problems still remain. Their research solely relies on either morphological operations or multi-scale feature extraction, without leveraging the complementarity between the two. Multi-scale feature extraction can capture both global and local information in images, allowing for the handling of more complex structures and enhancing the model's generalization ability. For the issues of difficult pericardial boundary segmentation and holes in the segmented images, morphological operations such as dilation and erosion can provide improvements. In this work, we propose a simple and effective method for accurate automatic detection of the pericardium and quantification of EATV from CCTA images. We introduce the Boundary-Enhanced Multi-scale U-Net network (BMT-UNet), which combines Boundary-Enhanced (BE), Multi-Scale (MS), and Convolutional Transformer (ConvT) modules. The network takes weighted multi-slice data as input to segment the pericardium, which addresses challenges in boundary segmentation difficulties and the presence of holes in the segmented images, enhancing the accuracy of EATV quantification. Specifically, first, we propose the BE module, which has the effect of enhancing the boundary features of the pericardial region through expansion and erosion operations, thus improving the accuracy of pericardial segmentation at the pericardial boundary. Furthermore, to accommodate variations in pericardial size in different slices, we propose the MS module. It introduces multi-scale convolution kernels within the same layer to capture the features of pericardial structures of different sizes and then fuse the obtained multi-scale information. In addition, given the large size of pericardial structures in the middle and base of the heart, it is particularly important in capturing global information and effective contextual relationships. Therefore, the ConvT module is introduced at the deepest layer of the network, which combines the advantages of convolution and Transformer to improve the global understanding of the pericardial structure. In addition, we propose a weighted multi-slice strategy that fully exploits the information of certain slices to preserve and capture the three-dimensional structure of the pericardium, and fill in the inside holes in the segmented pericardial image.

Due to the extensive use of CT scans for coronary artery calcium scoring, a large amount of CT scan data has been accumulated, providing a valuable resource for future research on EAT. Given the above-mentioned segmentation problems and the large amount of available training data, this study aims to develop a fully automatic and accurate method for quantifying EAT from CCTA images. The main contributions of this work are summarized as follows:

(1) The BE module is proposed to enhance the boundary features of the pericardial region through dilation and erosion operations, thereby improving the accuracy of pericardial boundary segmentation.
(2) The BMT-UNet network, which combines the BE module, multi-scale feature extraction, and ConvT module, has been designed to enhance the reliability of pericardial boundary segmentation.
(3) A weighted multi-slice strategy is proposed to fully leverage information from adjacent slices, retaining the three-dimensional structure of the pericardium. This strategy improves the accuracy of pericardial segmentation and EATV quantification, thereby facilitating better automatic detection of the pericardium and quantitative analysis of EATV.

2. Materials and methods

2.1. Dataset

The CT dataset used here was obtained from the General Hospital of North Theater Command. CCTA images of 50 patients were randomly selected. Among them, the training set contained 30 cases, and the validation set and test set each contained 10 cases. Each case consisted of 200–300 slices of size 512 $\times$ 512 pixels. As shown in Table 1, the patients ranged in age from 34 to 68 years old, of whom 27 (54 %) were female. The EATV obtained by manual segmentation (i.e. taken as the ground truth) ranged from 41.0 to 252.9 ml, with a mean value of 118.9 ml. To ensure accuracy, the data annotation was conducted by members of the laboratory research team under the guidance of radiologists (Y.S. and B.Y). The annotated results were reviewed and verified by the radiologists, and standards were jointly developed.

2.2. Method overview

The pericardium is a fibrous serous sac that envelops the heart and the root of the aorta, as shown in Fig. 1. Its outer layer is the fibrous pericardium, which is thick and tough, while the inner layer is the serous pericardium. The serous pericardium is further divided into the visceral and parietal layers. The visceral layer is closely attached to the outer surface of the myocardium, while the parietal layer is closely attached to the inner surface of the fibrous pericardium. The space between the two layers forms a closed cavity called the pericardial cavity. In CCTA images, the pericardium appears as a thin white line (Fig. 2(a)), and the label of the pericardium is shown in Fig. 2(b). Surrounding the pericardium, is a variable amount of adipose tissue. We focus on the EAT that is contained within the pericardium (Fig. 2(c), (d)). The overall structure of which is shown in Fig. 3. First, a series of slices are fed into a deep neural network model to obtain the segmentation of the pericardium. Subsequently, the adipose tissue within the pericardium, namely the EAT, is automatically quantified using standard fat attenuation ranges.

Table 1
Descriptive statistics of the patients manually measured EAT and ages [25].

	EATV (ml)	Age (years)
Minimum	41.0	34
Maximum	252.9	68
Mean	118.9	53.3
Std.Deviation	42.2	8.7

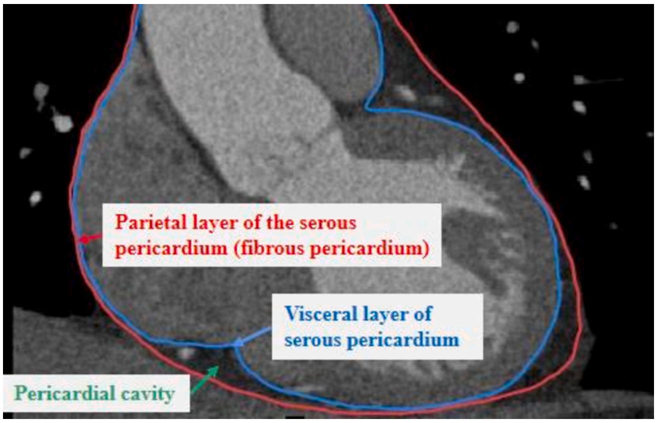

Fig. 1. Structure of the pericardium.

2.3. Pericardial segmentation

In this section, we describe in detail the methods used for pericardial segmentation with the aim of improving the accuracy and stability of pericardial segmentation. This section is divided into three subsections. In section 2.3.1, we present the BMT-UNet network for pericardial segmentation. In section 2.3.2, we describe the weighted multi-slice strategy employed in the BMT-UNet network. Finally, in section 2.3.3, we introduce the loss function used in this study.

2.3.1. Boundary-enhanced multi-scale U-Net with convolution Transformer (BMT-UNet)

The structure of the pericardium segmentation network BMT-UNet we propose is based on U-Net. Our aim is to better handle variations in pericardium structure size, smooth the boundaries of the pericardial segmentation results, and fill in the holes within the pericardial segmentation results, so as to obtain more complete and accurate pericardial segmentation results. Fig. 4 shows the structure of the BMT-UNet network.

Compared to the traditional U-Net, the BMT-UNet uses multi-scale blocks instead of the basic convolutional blocks. The input data consists of the current slice and the two nearest contiguous slices on each side. Among these five slices, each is assigned a corresponding weight, namely w_1, w_2, w_3, w_2, and w_1. These slices are concatenated along the channel dimension, resulting in an input data size of $5 \times H \times W$, where H and W represent the height and width of the input image. The encoder path consists of four encoding layers, each composed of a BE module and a MS module. Each layer is followed by a max pooling operation with a stride size of 2 to downsample the feature maps. The last layer of the encoder incorporates a ConvT module to enhance the exploitation of global contextual information [36]. In the decoder stage, there are four upsampling layers. Each upsampling layer consists of an upsampling block (including a transposed convolution with a stride of 2, a 3x3 convolution, a batch normalization, a PReLU activation function) and a MS module. The feature maps obtained from the encoder stage and decoder stage are concatenated through skip connections, combining low-level and high-level features to provide richer semantic information and details, prompting the network to recover detailed structure and improve localization accuracy.

Multi-scale module. In medical image segmentation, there is a need to accurately extract ROIs, which may have large variations in size. To address this problem, many researchers have proposed to use multi-scale modules to handle the features at different scales [37–40]. By employing these modules, information from target regions can be effectively captured at various resolution levels. In this study, within the sequence of slices from the apex to the base (i.e., from the upper portion of the slice sequence to the lower portion), the size of the pericardium in the ROI typically varies with the slice position. The range of variation is roughly from a few centimeters to several tens of centimeters. Inspired by this, we propose a multi-scale module to cope with the problem of drastically varied areas of the pericardium in different slices so as to effectively capture information from pericardial regions with different sizes. As shown in Fig. 4, we use multi-scale modules with dilated convolutions to replace the basic convolutional blocks in the original U-Net.

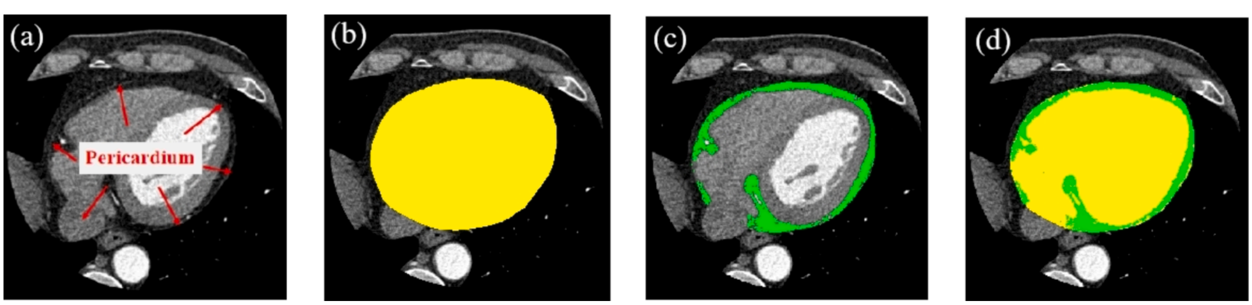

Fig. 2. Labelling the pericardium and EAT. (a) is the original CT image, (b) shows the labelled pericardium and its inner region in yellow, (c) shows the labelled EAT in green, and (d) is the superposition of the two labelled regions (b) and (c).

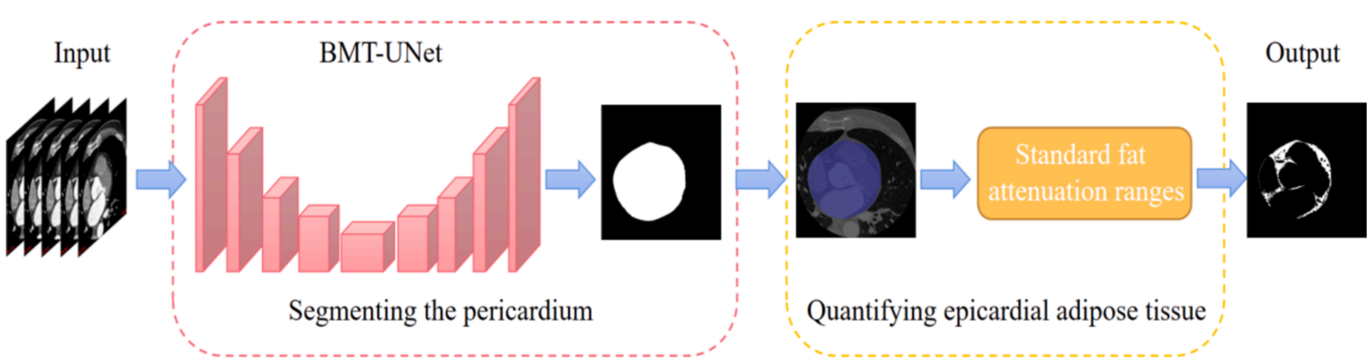

Fig. 3. Overall flowchart of the proposed method.

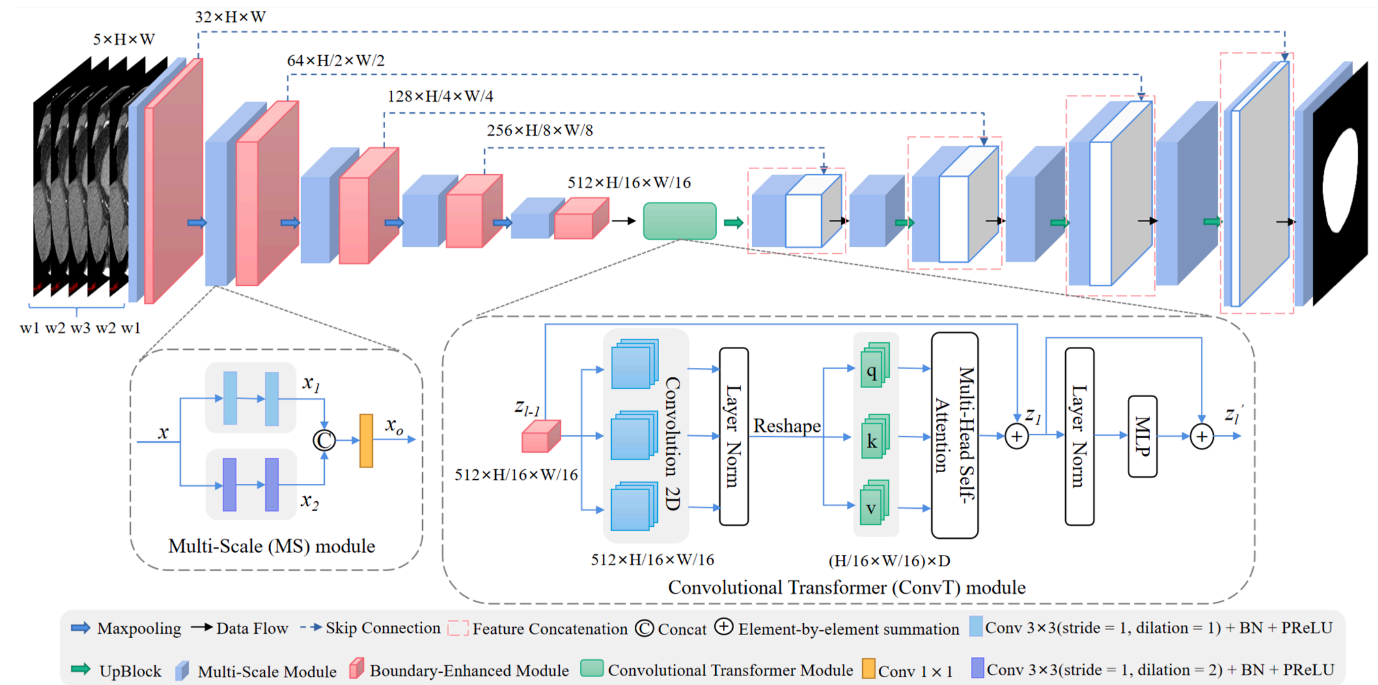

Fig. 4. Architecture of BMT-UNet.

Although the U-Net encoder captures multi-scale features at multiple stages, pericardial structures of different sizes still exist within the same receptive field. To better cope with this size variation, especially within the same level, we introduce an intra-layer multi-scale strategy. The MS module contains two convolutional blocks with different receptive fields, considering multi-scale information in parallel. Feature fusion is achieved by concatenating the feature maps of the two convolutional blocks to complement the information of different receptive fields and provide a richer and more comprehensive feature representation. This multi-scale information fusion strategy not only adapts to smaller regions of interest, but also handles larger regions. Specifically, the MS module consists of two branches, with each branch containing two 3×3 convolutional blocks. Each convolutional block consists of a dilated convolution with different dilation rate (rate $= 1$ or rate $= 2$), followed by batch normalization and a PReLU activation function. x represents the input features, x_1 and x_2 represent the features obtained from the convolutional blocks with different dilation rates, and x_o represents the output features of the multi-scale convolutional block. The expressions for calculating the output features x_o are as follows:

$$x_1 = w_2(w_1 x + b_1) + b_2 \tag{1}$$

$$x_2 = w_{r2}(w_{r1} x + b_{r1}) + b_{r2} \tag{2}$$

$$x_o = w_o(Cat[x_1, x_2]) + b_o \tag{3}$$

where w_1, w_2, w_{r1}, w_{r2} and w_o represent the weight of the convolutional kernel. b_1, b_2, b_{r1}, b_{r2} and b_o represent the bias of the convolutional kernel.

Boundary-enhanced module. The BE module aims to enhance the model's perception of pericardial shape and boundary by fusing the enhanced boundary information with the original feature maps. Fig. 5 illustrates the specific structure. First, we use a sigmoid function to map the input feature map y to the range from 0 to 1, ensuring the stability of subsequent operations. Then, we perform dilation and erosion operations respectively on the mapped results to obtain feature maps y1 and y2. The dilation operation increases the feature intensity around the boundaries of pericardium, thus helping the model to better detect and segment the boundary of the pericardium. By reducing the pixel values around the boundary regions, the erosion operation has the effect of weakening irregular edges while further emphasizing the shape information of the pericardium. Next, we subtract the eroded feature map y2 from the dilated feature map y1 to highlight the boundary of the pericardium and obtain a feature map y3 that enhances the boundary information. The feature map y3 can more accurately locate the boundary and thus can provide more accurate shape information for subsequent tasks. Finally, we perform element-wise multiplication on the original

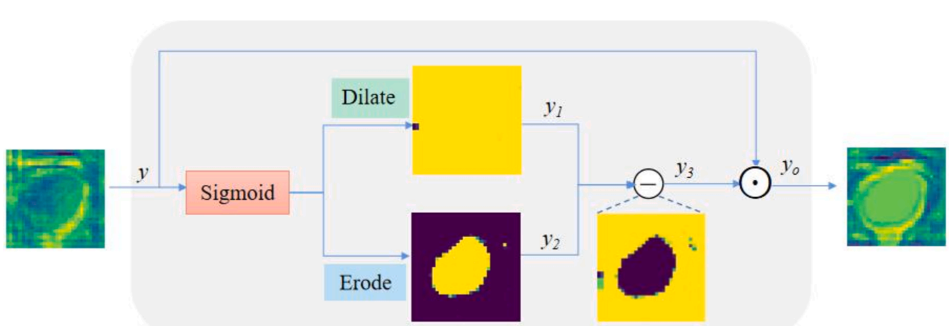

Fig. 5. Structure of the Boundary Enhancement (BE) module.

feature map y and the enhanced boundary information feature map y3 to obtain a fused feature map y_o. As the feature map y_o integrates both the original features and the enhanced boundary information, it focuses more on the important features around the boundary region while also retaining the global contextual information of the original features. This fusion process further enhances the model's perception of the pericardial shape and boundary, thus providing more accurate and reliable input for pericardial detection and segmentation. Formally, the output feature y_o of the BE module are calculated as follows:

$$y_1 = \mathrm{Dilate}(\mathrm{Sigmoid}(y)) \tag{4}$$

$$y_2 = \mathrm{Erode}(\mathrm{Sigmoid}(y)) \tag{5}$$

$$y_3 = y_1 - y_2 \tag{6}$$

$$y_o = y \odot y_3 \tag{7}$$

Convolutional Transformer module. To better preserve the spatial context information in the feature map as well as to capture dependencies over longer distances, we propose the Convolutional Transformer module, as shown in Fig. 4. It is worth noting that we only use the ConvT module in the last layer of the encoder, which can save computational resources while maximize the utilization of the extracted high-level features for global context integration and feature fusion. In the traditional Transformer module, q, k and v (queries, keys and values, respectively) are computed through linear transformations. However, in the ConvT module, the linear layer in the traditional Transformer module is replaced with a 2D convolutional layer to compute q, k, and v. Compared with the linear projection, the convolution operation preserves the spatial information within the local receptive field and can make better use of the spatial structure features of the pericardial images. In addition, the convolution operation can utilize feature interactions within local regions to capture more extensive contextual information, which is particularly important in pericardial segmentation because holes within the segmented pericardial image are usually caused by the lack of contextual information. Therefore, through the ConvT module, we can effectively fill the holes and improve the accuracy of the segmentation. Specifically, the input features are first entered into a convolutional mapping (including a 3 × 3 convolution, ReLU activation function, and LayerNorm) with output dimension D to obtain q, k, and v. Then they are reshaped as $H/16 \times W/16 \times D$. Next, they are fed into the MHSA module [41]. The calculation process of the ConvT module is as follows:

$$z_l = \mathrm{MHSA}(\mathrm{Flatten}(\mathrm{LN}(\mathrm{Conv}(z_{l-1})))) + z_{l-1} \tag{8}$$

$$z'_l = (\mathrm{MLP}(\mathrm{LN}(z_l))) + z_l \tag{9}$$

where z_l and z'_l are the outputs of the MHSA module and the MLP module in the lth layer. Conv denotes the convolution operation and LN denotes layer normalization.

2.3.2. The weighted multi-slice strategy

Although many CNN based methods have improved the accuracy of automated EAT segmentation, there is often a trade-off between the use of 2D spatial receptive fields and inter-slice information due to the limitation of GPU memory when using 2D CNN or 3D CNN for segmentation [42,43]. On the one hand, 2D CNNs only consider the information from individual slices, ignoring the spatial continuity between slices. On the other hand, 3D CNNs such as 3D U-Net [44] and V-Net take into account the three-dimensional spatial information of the image and can process multiple slices simultaneously. However, due to the more demanding memory and computational requirements [45], the receptive field is always limited. The limited receptive field can increase the difficulty of distinguishing the pericardium from surrounding similar tissues, making it challenging to identify the pericardial boundary and prone to segmentation errors. To address these problems, this study combines the advantages of both 2D and 3D CNNs by proposing a weighted multi-slice strategy. The use of weighted multi-slice data as input in the BMT-UNet network can further enhance the global understanding and segmentation ability of pericardial structure.

This weighted multi-slice strategy uses several continuous slices as input data and assigns different weights to each slice, based on the number of slices involved. The multi-slice strategy exploits the information from adjacent slices by considering the contextual relationship between them, thereby preserving and capturing the three-dimensional structural information of the pericardium in the image. Compared to the methods that segment EAT using a single slice alone, this strategy can better handle variations in the shape, size, and position of the ROI. The weighted multi-slice strategy emphasizes or reduces the importance of certain slices by assigning different weights to each slice. Specifically, the segmentation of slice i requires the use of information from slice $i-\lfloor c/2\rfloor$ to slice $i+\lfloor c/2\rfloor$. According to *a priori* knowledge, the maximum weight is assigned to slice i, followed by weights for slice i-1 and slice $i+1$, then slice i-2 and slice $i+2$, and so on, and the sum of the weights is equal to 1. If $i < \lfloor c/2\rfloor + 1$, the first slice is copied $\lfloor c/2\rfloor - i + 1$ times and if $i > n - \lfloor c/2\rfloor$, the last slice is copied $\lfloor c/2\rfloor - n + i$ times. n represents the total number of slices in the volume data, and c denotes the number of input slices. The input data can be considered as a 2D image with c image channels, where each slice occupies one channel, and the number of input channels is c. Using this weighted multi-slice strategy allows a 2D network to process 3D information, improving the accuracy of EAT segmentation. The weighted multi-slice method divides 3D data into multiple 2D slices, which have lower dimensions, resulting in reduced computational and storage overhead. Meanwhile the weighted multi-slice method can process multiple slices in parallel, thereby accelerating the computational process.

2.3.3. Loss function

Since pericardial tissues may be less relative to other tissues in CCTA images, the pericardial segmentation task suffers from the class imbalance problem. In this study, Dice Loss, which is suitable for the segmentation of the foreground- background imbalance problem, is used to avoid the model being biased towards the background classes. In addition, the computation of Dice Loss considers the similarity between two sets and focuses on the overlapping region, which reduces the effect of individual pixels on the overall loss. This makes the model more stable and avoids drastic changes in gradient when optimizing in the boundary region. The calculation of Dice Loss is based on the Dice Similarity Coefficient (DSC), which is calculated as follows:

$$Dice\,Loss = 1 - \frac{2TP}{2TP + FN + FP} \tag{10}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

2.4. EAT segmentation and quantization

In this study, combined with advice of two of us with clinical experience (Y.S and B.Y.), the reference standard for the threshold range of EAT was determined as −190 HU to −30 HU. This range corresponds to the standard attenuation range for fat tissue in CT and has been validated in previous studies by other researchers [24,46–48]. After pericardial segmentation, pixels within the range of −190 to −30 HU are taken as EAT to obtain the quantification results of EATV.

3. Experiments and results

3.1. Experimental details

All experiments were completed on a NVIDIA 3090 GeForce graphics card with 24 GB memory. And all processing was implemented in the

PyTorch framework, which provides a rich set of tools and functions and facilitates rapid and efficient training of the models. To optimize the network training process, we adjusted the network parameters with the Adam optimizer to ensure the network converge more quickly and hence increase the training speed. In the training process, we set the batch size to 4 and conducted training for 100 epochs. The initial learning rate was set to 10^{-4}, and after the 60th epoch, the learning rate was decayed by 2.5×10^{-6} per epoch. The number of attention heads in the ConvT module was set to 8.

3.2. Evaluation metrics

We used five evaluation metrics to assess the effectiveness of the new and comparison methods: Dice Similarity Coefficient (DSC), Recall, Precision, Hausdorff Distance (HD), and Average Symmetric Surface Distance (ASSD). The purpose and defining expressions of these metrics are as follows:

(1) Dice Similarity Coefficient (DSC)

DSC describes the similarity between the segmentation result and the Ground Truth. It is calculated as

$$DSC = \frac{2TP}{2TP + FN + FP} \tag{11}$$

(2) Recall

Recall measures the ratio of correctly predicted positive samples in the segmentation result to the actual positive samples, and is computed as

$$Recall = \frac{TP}{TP + FN} \tag{12}$$

(3) Precision

Precision, also known as positive predictive value, reflects the proportion of the number of positive samples classified by the model that are correctly classified. It is defined as:

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

(4) Hausdorff Distance (HD)

HD is the maximum distance from a point in one set to the nearest point in another set and describes the degree of similarity between two sets of points, with a smaller value showing a higher segmentation accuracy. It is calculated as

$$HD(A, B) = \max[(\min\|a - b\|), \max(\min\|b - a\|)] \tag{14}$$

where a represents a point in target region A in the segmentation result, and b represents a point in the target region B of the ground truth.

(5) Average Symmetric Surface Distance (ASSD)

ASSD measures the average symmetrical surface distance between the predicted segmentation boundary and the ground truth segmentation boundary. The smaller the *ASSD*, the better the segmentation result. It is defined as:

$$ASSD(A, B) = \frac{\sum d(a, B) + \sum d(b, A)}{(|A| + |B|)} \tag{15}$$

where $d(a,B)$ refers to the shortest distance from a to B, and $d(b, A)$ likewise.

3.3. Ablation study

In order to validate the effectiveness of the proposed MS module, BE module, and ConvT module, as well as analyze their respective contributions to the network performance, ablation experiments were conducted and are described in this section. Tables 2 and 3 present the results. The U-Net serves as the baseline model, and the other models incorporate the corresponding modules on top of U-Net.

The results show that adding these three modules separately can improve recall in the pericardial segmentation task, indicating that more pericardial regions are correctly segmented, and subsequently improving the accuracy of EAT recognition in the second stage. It can be seen from Table 3 that the MS module and BE module play an important role in improving the Recall, while the ConvT module has a substantial impact in reducing false positive regions and improving the Precision. By combining the advantages of the MS, BE, and ConvT modules, BMT-UNet achieves outstanding performance in terms of DSC, Recall, Precision, ASSD, and HD. The DSC for EAT segmentation improves by 1.14 %.

3.4. Impact of the number of input slices

Based on the BMT-UNet network structure shown in Fig. 4, we tested the effect of 3 and 5 slices on the five evaluation metrics and compared these to the single slice case. The results are shown in Table 4. We have experimented with 7 slices, but the effect is not as good as 3 and 5 slices, which are not shown here for simplicity.

This result demonstrates that (Table 4), compared to single slice input data, the multi-slice input leads to an improvement in four of the evaluation metrics (DSC, Precision, HD and ASSD), but a slight decrease in *Recall*. This may be due to the fact that as the number of slices increases, more contextual information is available in the input data, and the network is more likely to focus on details and local features, thus improving the accuracy of segmentation. However, when considering multiple slices, the variations in local structures may result in inconsistent features of certain EAT parts across multiple slices, which could make it difficult for the model to accurately segment them, leading to a decrease in recall. To solve this problem, we propose a solution, which is to assign appropriate weights to each slice, so that the network can more selectively utilize the information of different slices. For a given number of slices ($c = 3$ or $c = 5$), the weighted multi-slice strategy ($Wc = 3$ or $Wc = 5$) outperforms its unweighted counterpart ($c = 3$ or $c = 5$) in terms of *DSC* and *Recall*. This indicates that by adjusting the weights of the slices, we can better utilize the correlation in the information between adjoining slices.

3.5. Comparison with existing methods

In order to further evaluate the proposed method, we compared it with several classic deep learning medical image segmentation approaches, including 2D U-Net[18], Attention UNet [49], UNet++ [50], SegNet [51], DilaLab [52], DAE-Former [53], V-Net and 3D U-Net[44]. Based on the open-source code, we retrained each of the five networks on the same training set as our model. Table 5 shows a qualitative comparison of the proposed method with other methods for the pericardial segmentation. Table 6 presents a qualitative comparison of the proposed method with other methods for the EAT segmentation. The proposed method achieved better results in all the evaluation metrics than those of the other segmentation models.

During the segmentation process, we indeed observed that the EAT around the apex and base positions is more difficult to segment compared to the middle region of the heart, being prone to under- or over-segmentation. To demonstrate the superiority of the proposed method more intuitively, Fig. 6 compares the visualization results of different networks for pericardial and EAT segmentation at the apex, the middle region of the heart, and the base. It is evident that our method is less prone to under- and over-segmentation, regardless of whether it is

Table 2
Results of module ablation experiments for pericardium segmentation. The symbol "√" indicates the inclusion of the corresponding module in the model. In the top row of this and subsequent tables, up arrows indicate that an increase in the evaluation metric implies better performance, the reverse being the case for the down arrows. The best value for each evaluation metric is shown in bold.

Method	BE	MS	ConvT	DSC↑	Recall↑	Precision↑	HD↓	ASSD↓
U-Net				0.978 ± 0.003	0.983 ± 0.01	0.974 ± 0.007	9.9 ± 3.2	2.4 ± 0.4
BE-UNet	√			0.979 ± 0.003	0.985 ± 0.009	0.973 ± 0.007	10.3 ± 2.8	2.6 ± 0.4
MS-UNet		√		0.980 ± 0.002	0.986 ± 0.008	0.974 ± 0.006	7.3 ± 1.9	1.9 ± 0.2
ConvT-UNet			√	0.979 ± 0.003	0.985 ± 0.009	0.975 ± 0.007	9.1 ± 3.2	2.3 ± 0.4
BMT-UNet	√	√	√	**0.981 ± 0.002**	**0.987 ± 0.007**	**0.975 ± 0.006**	**7.0 ± 1.2**	**1.8 ± 0.2**

Table 3
Results of module ablation experiments for EAT Segmentation. The symbol "√" indicates the inclusion of the corresponding module in the model.

Method	BE	MS	ConvT	DSC↑	Recall↑	Precision↑	HD↓	ASSD↓
U-Net				0.924 ± 0.020	0.942 ± 0.019	0.908 ± 0.031	2.6 ± 0.5	0.8 ± 0.1
BE-UNet	√			0.931 ± 0.019	0.948 ± 0.022	0.914 ± 0.033	2.5 ± 0.7	0.5 ± 0.1
MS-UNet		√		0.932 ± 0.018	0.948 ± 0.022	0.917 ± 0.035	2.3 ± 0.5	**0.4 ± 0.1**
ConvT-UNet			√	0.931 ± 0.017	0.946 ± 0.022	0.919 ± 0.034	2.4 ± 0.7	**0.4 ± 0.1**
BMT-UNet	√	√	√	**0.936 ± 0.017**	**0.950 ± 0.021**	**0.922 ± 0.030**	**2.2 ± 0.6**	**0.4 ± 0.1**

Table 4
The effect of using different number of input slices in pericardium and EAT segmentation. $Wc = 3$ indicates that three slices were weighted, where the weights were 0.3, 0.4 and 0.3, respectively (these being selected on the basis of previous experience). $Wc = 5$ indicates five weighted slices, with values of 0.1, 0.2, 0.4, 0.2, and 0.1 assigned to the five slices, respectively.

	Pericardium		EAT				
	DSC↑	HD↓	DSC↑	Recall↑	Precision↑	HD↓	ASSD↓
$c = 1$	0.981 ± 0.002	7.0 ± 1.2	0.936 ± 0.017	0.950 ± 0.021	0.922 ± 0.030	2.2 ± 0.6	0.4 ± 0.1
$c = 3$	0.982 ± 0.002	**5.7 ± 0.6**	0.938 ± 0.015	0.939 ± 0.023	**0.938 ± 0.025**	2.1 ± 0.5	0.4 ± 0.1
$Wc = 3$	0.982 ± 0.002	6.1 ± 0.8	0.938 ± 0.017	0.949 ± 0.023	0.928 ± 0.026	2.1 ± 0.5	0.4 ± 0.1
$c = 5$	0.982 ± 0.003	5.8 ± 0.9	0.937 ± 0.018	0.947 ± 0.024	0.928 ± 0.023	2.1 ± 0.5	**0.3 ± 0.1**
$Wc = 5$	**0.983 ± 0.002**	**5.7 ± 0.6**	**0.939 ± 0.017**	**0.953 ± 0.019**	0.926 ± 0.029	**2.1 ± 0.4**	0.4 ± 0.1

Table 5
The comparative experimental results for Pericardium segmentation. Optimal values among all eight methods are shown in bold.

	PDice↑	PHD↓	Params	FLOPs
2D U-Net [18]	0.978 ± 0.003	9.9 ± 3.2	34.53 M	261.92 G
Atten-UNet [49]	0.978 ± 0.003	10.8 ± 5.6	35.93 M	279.53 G
UNet++ [50]	0.979 ± 0.004	8.5 ± 3.2	36.63 M	554.34 G
SegNet [51]	0.978 ± 0.004	9.5 ± 2.9	**4.71 M**	29.35 G
DilaLab [52]	0.978 ± 0.005	8.0 ± 2.2	20.46 M	**23.02 G**
V-Net [20]	0.966 ± 0.003	7.6 ± 1.5	57.88 M	378.14 G
DAE-Former [53]	0.966 ± 0.003	11.8 ± 2.3	7.52 M	34.73 G
3D U-Net [44]	0.971 ± 0.003	8.8 ± 3.2	51.72 M	421.34 G
Proposed	**0.983 ± 0.002**	**5.7 ± 0.6**	14.07 M	85.61 G

*PDice: Pericardium Dice; PHD: Pericardium HD (mm).

the apex, base, or middle region of the heart.

A plot of the individual EAT estimates for the 10 test cases obtained by the method proposed here, against the corresponding ground truth values (Fig. 7), gave a Pearson correlation coefficient (r) of 0.99 ($P < 0.001$) with a gradient of 0.98, indicating a strong correlation between the two methods. Bland-Altman analysis (Fig. 8) showed that all but one of the points lay within the 95 % confidence intervals of the mean difference. The deviation (green line) was close to zero.

Table 6
The comparative experimental results for EAT segmentation. Optimal values among all eight methods are shown in bold.

	MAVD↓	BAB↓	CCC↑	PC↑	EDice↑	EHD↓
2D U-Net [18]	4.64	4.10	0.985	0.990	0.924 ± 0.020	2.6 ± 0.5
Atten-UNet [49]	5.74	4.45	0.979	0.990	0.922 ± 0.017	2.6 ± 0.7
UNet++ [50]	4.58	2.10	0.983	0.984	0.929 ± 0.018	2.4 ± 0.4
SegNet [51]	5.70	4.40	0.977	0.986	0.926 ± 0.017	2.6 ± 0.4
DilaLab [52]	6.49	4.90	0.975	0.987	0.925 ± 0.020	2.7 ± 0.5
V-Net [20]	4.23	3.70	0.939	0.944	0.923 ± 0.016	3.1 ± 0.5
DAE-Former [53]	5.81	4.85	0.957	0.962	0.917 ± 0.017	5.2 ± 0.5
3D U-Net [44]	4.27	3.85	0.971	0.975	0.926 ± 0.018	3.8 ± 0.4
Proposed	**2.01**	**1.10**	**0.992**	**0.993**	**0.939 ± 0.017**	**2.1 ± 0.4**

*MAVD: mean absolute volume difference (ml); BAB: Bland-Altman Bias; CCC: concordance correlation coefficient; PC: Pearson correlation; EDice: EAT Dice; EHD: EAT HD (mm).

4. Discussion

As EAT is located within the pericardium (see Fig. 2), accurate segmentation of the pericardium becomes a crucial step. We propose a weighted multi-slice strategy and the BMT-UNet that combines Boundary-Enhanced, Multi-Scale, and Convolutional Transformer modules, increasing the model's ability to perceive the shapes and edges of pericardial structures at different scales, as well as capturing local image features and long-distance dependencies in the image, thereby better handling the challenges of accurate pericardial boundary segmentation and the holes presented within the segmented region. In addition, the weighted multi-slice strategy can make better use of the contextual information between slices, thus further improving the accuracy of pericardial segmentation. After segmenting the pericardium, EAT quantification is accomplished through thresholding, i.e. pixels with CT values within −190 HU and −30 HU are taken as EAT. Table 5 indicates that our method achieves the best segmentation performance

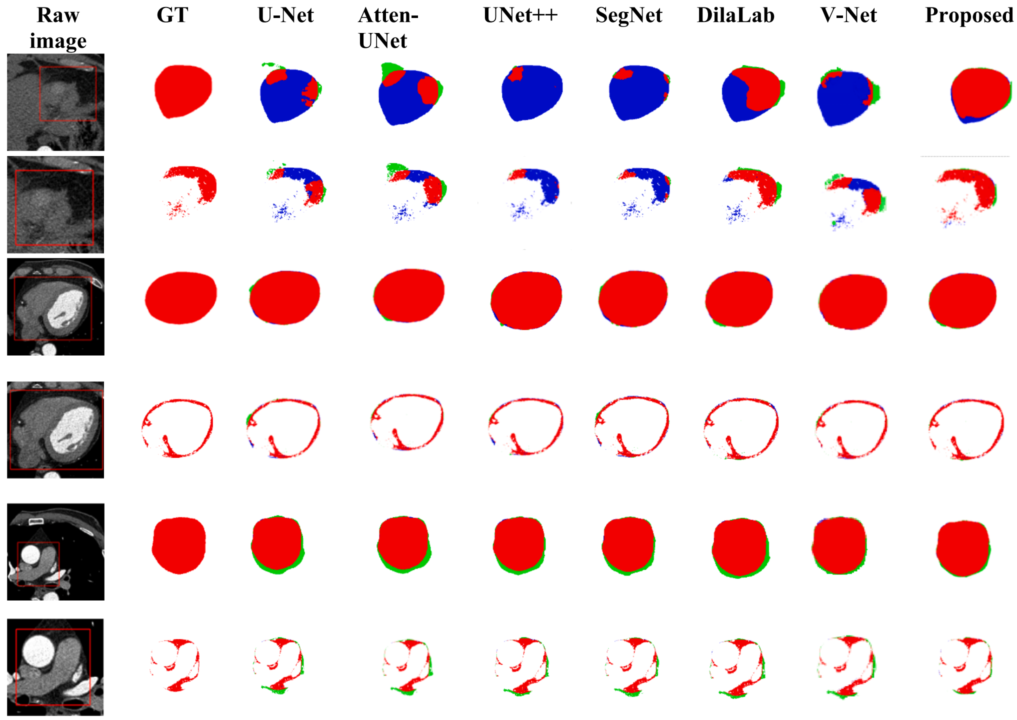

Fig. 6. Visualization of pericardial (rows 1, 3, and 5) and EAT (rows 2, 4, and 6) segmentation results by different networks at the apex (top two rows), the middle region (middle two rows), and the base (bottom two rows) of the heart. Green, blue and red represent over-segmentation (false positives), under-segmentation (false negatives) and the correctly segmented areas, respectively.

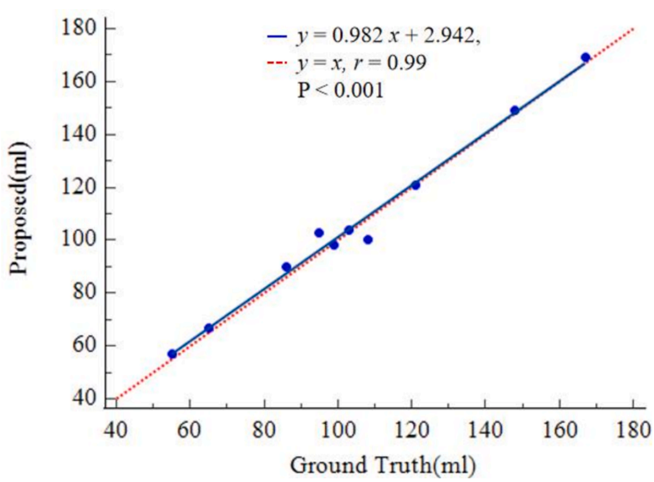

Fig. 7. Regression plot of EATV.

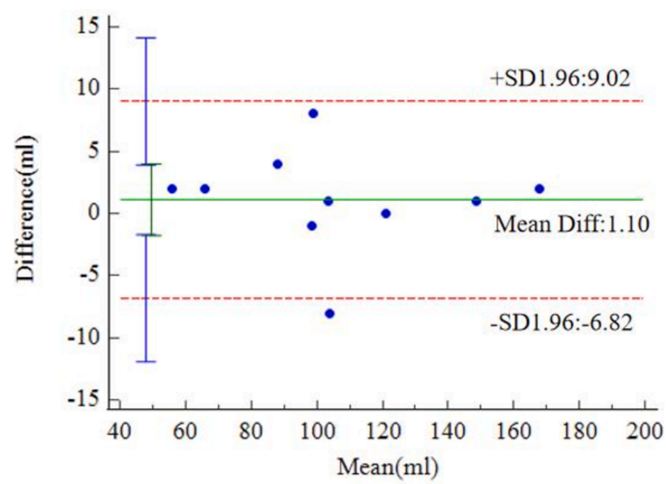

Fig. 8. Bland–Altman analysis. The green solid line is the mean of the differences between the two methods and the red dashed lines are the upper and lower limits of the 95% confidence interval.

in the pericardium segmentation task while maintaining relatively low computational costs and parameter requirements. Meanwhile, Table 6 shows that the mean absolute volume difference of the quantified EATV between our method and the baseline network U-Net is 2.91 and 4.64 ml, respectively. Our method also results in a smaller measurement error in EATV. Compared to other classical segmentation networks such as U-Net, Attention UNet, UNet++ and V-Net, our approach achieves higher accuracy and generates fewer cases of under- and over-segmentation, as shown in Fig. 6. Better pericardial segmentation result can lead to improved EAT segmentation and quantification accuracy. As shown in Fig. 7, the correlation coefficient between the EATV computed using our method and the ground truth is close to 1, and it can be seen from Fig. 8 that the absolute difference between the quantified EATV using our algorithm and the actual volume is relatively small, both results indicating that our method can accurately determine the volume of EAT. However, this study still has some limitations First, deep learning methods usually require a large dataset for learning more diverse features. Therefore, an epicardial adipose tissue dataset containing a large number of subjects' cases is crucial for training the model more thoroughly. Future research will investigate larger and more diverse data sets, including those covering more manufacturers and more models of CT scanning equipment. Secondly, as can be seen from the evaluation metrics (Table 6) and visualization (Fig. 6) of the EAT segmentation results, our method generates some obvious false-positive errors near the boundary of the pericardium. This indicates that although the boundary-enhancement module is effective in making the segmented boundary more accurate, it still has room for improvement. In our future work, we will further consider using boundary loss and constraining the segmented boundaries of pericardium to be smooth across nearby slices to tackle this issue.

5. Conclusion

In this paper, we have proposed a deep learning method for automatically segmenting the pericardium and quantifying EAT from CCTA images. The BMT-UNet network innovatively integrates Boundary-Enhanced, Multi-Scale, and Convolutional Transformer modules to enhance boundary clarity, address size variations, and fill the holes in the segmented pericardium for precise pericardium segmentation. The method provides a powerful tool for clinicians and researchers, allowing for the rapid and autonomous quantification of EATV in patients. Experimental results demonstrate that our method outperforms classical deep learning models on the CCTA dataset, offering superior DSC and HD metrics while maintaining a relatively low computational load and parameter count. Furthermore, visualization results indicate a strong concordance between the quantified EATV and the ground truth. In summary, the proposed method significantly enhances EAT quantification efficiency and accuracy, holding promise for future clinical applications.

CRediT authorship contribution statement

Ying Wang: Writing – original draft, Methodology, Conceptualization. **Ankang Wang:** Writing – review & editing, Methodology, Conceptualization. **Lu Wang:** Writing – review & editing, Conceptualization. **Wenjun Tan:** Writing – review & editing. **Lisheng Xu:** Writing – review & editing, Funding acquisition. **Jinsong Wang:** Methodology, Investigation. **Songang Li:** Writing – review & editing. **Jinshuai Liu:** Writing – review & editing. **Yu Sun:** Data curation. **Benqiang Yang:** Data curation. **Steve Greenwald:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by the National Natural Science Foundation of China (No. 62273082, No. U21A20487), the Liaoning Province Science and Technology Plan Project (No. 2023JH2/101300125), the Natural Science Foundation of Liaoning Province (Grant 2021-YGJC-14), the Basic Scientific Research Project (Key Project) of The Educational Department of Liaoning Province, (LJKZ00042021), the Fundamental Research Funds for the Central Universities (No. N2319001 and No. N2119008) and the Shenyang Science and Technology Plan Fund (No. 21-104-1-24).

Data availability

The authors do not have permission to share data.

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